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FRESH-KEEPING BAG

Abstract

A fresh-keeping bag includes a body, a mouth is defined at a top of the body, a first sealing strip and a second sealing strip are respectively provided on two sides of the mouth, the first sealing strip is provided with a snap-in protrusion, a snap-in groove is defined on the second sealing strip at a position corresponding to the snap-in protrusion, transverse protrusions are provided between two ends of the snap-in protrusion and a bottom of the mouth, the transverse protrusions extend horizontally relative to the snap-in protrusion, transverse grooves corresponding to the transverse protrusions are formed between two ends of the snap-in groove and the bottom of the mouth, and the transverse protrusions are engaged with the transverse grooves after the snap-in protrusion is engaged with the snap-in groove, to completely seal the mouth.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application is a continuation of PCT application Ser. No. PCT/CN2023/082236, filed on Mar. 17, 2023, which claims the priority and benefit of Chinese patent application serial no. 202211444322.6, filed on Nov. 18, 2022. The entireties of PCT application Ser. No. PCT/CN2023/082236 and Chinese patent application serial no. 202211444322.6 are hereby incorporated by reference herein and made a part of this specification.

TECHNICAL FIELD

[0002] The present disclosure relates to the field of packaging bags, and in particular, to a fresh-keeping bag.

BACKGROUND ART

[0003] In modern family life, vegetables, fruits, and other foods are usually stored and preserved in environments such as refrigerators that can maintain a lower temperature. In order to reduce dehydration of fresh foods such as vegetables and fruits, they are often sealed with fresh-keeping bags. Of course, it is also possible to seal it with fresh-keeping bags to prevent flavor transfer.

[0004] Referring to FIGS. 1 and 2, an existing fresh-keeping bag has a mouth 13 at a top of a body 10 of the fresh-keeping bag. In order to seal the mouth 13, a first sealing strip 131 and a second sealing strip 132 are provided on two sides of the mouth 13, respectively, a snap-in protrusion 1311 is provided on the first sealing strip 131, a snap-in groove 1321 is defined at a position on the second sealing strip 132 corresponding to the snap-in protrusion 1311, the snap-in protrusion 1311 can be clamped into the snap-in groove 1321 to preliminarily seal the mouth 13. In use, users pinch the first sealing strip 131 and the second sealing strip 132 along an extension direction of the first sealing strip 131 and the second sealing strip 132 with their hands, so that the snap-in protrusion 1311 is pressed into the snap-in groove 1321, thereby quickly operating while ensuring good sealing effect. In order to enhance the seal of the mouth 13 at two ends of the first sealing strip 131 and the second sealing strip 132, second sealing protrusions 1314 are provided at the two ends of the first sealing strip 131 on an inner side thereof, and second sealing grooves 1324 are formed on the two ends of the second sealing strip 132 on an inner side thereof at positions corresponding to the second sealing protrusions 1314. The second sealing protrusions 1314 can be clamped in the second sealing grooves 1324, thereby further improving the seal of the mouth 13.

[0005] In the existing fresh-keeping bag mentioned above, the second sealing protrusions 1314 and the second sealing grooves 1324 extend along an extension direction of the snap-in protrusion 1311 and the snap-in groove 1321, which have no substantial difference from the snap-in protrusion 1311 and the snap-in groove 1321, making it difficult to achieve complete sealing. In addition, in an existing production process, a side opening is molded at the bag mouth, and then the side of the bag mouth is subjected to a secondary manual adhesive process and then is solidified by mold heating, and finally undergoes a secondary vulcanization treatment. An existing processing and production mentioned above require four processes: mold forming, manual adhesive processing, solidifying by mold heating and a secondary vulcanization. In terms of product molding process, a two-stage process of secondary manual adhesive processing and mold solidifying by heating solidification must be adopted. An increase in existing production processes leads to increased

costs and decreased efficiency, while also prone to a decline in product quality stability, such as tearing at the bag opening.

SUMMARY

[0006] The present application provides a fresh-keeping bag that solves the problems of poor sealing effect and complex production process caused by a second sealing protrusion and a second sealing groove extending along an extension direction of a protrusion and a snap-in groove in an existing fresh-keeping bag.

[0007] The present application solves the above technical problems through a fresh-keeping bag including a body, a mouth is defined at a top of the body, a first sealing strip and a second sealing strip are respectively provided on two sides of the mouth, the first sealing strip is provided with a snap-in protrusion, a snap-in groove is defined on the second sealing strip at a position corresponding to the snap-in protrusion, the mouth is preliminarily sealed after the snap-in protrusion is engaged with the snap-in groove, transverse protrusions are provided between two ends of the snap-in protrusion and a bottom of the mouth, the transverse protrusions extend horizontally relative to the snap-in protrusion, transverse grooves corresponding to the transverse protrusions are formed between two ends of the snap-in groove and the bottom of the mouth, and the transverse protrusions are engaged with the transverse grooves after the snap-in protrusion is engaged with the snap-in groove, to completely seal the mouth.

[0008] Each of the transverse grooves is designed as non-through in a thickness direction of sealing strips, the sealing strips respectively form groove ends at two ends of each of the transverse grooves, and the groove ends are completely engaged with two ends of each of the transverse protrusions after the transverse protrusions are engaged with the transverse grooves, to completely seal the mouth.

[0009] The transverse protrusions are semi-cylindrical in shape, a first side of each of the transverse protrusions gradually rises from the bottom of the mouth, a second side of each of the transverse protrusions is connected with a curved transition to the snap-in protrusion, and a top of each of the transverse protrusions is higher than a top of the snap-in protrusion.

[0010] A length of each of the transverse protrusions is greater than a thickness of the snap-in protrusion, and two of the transverse protrusions form a curved “I” in shape with the snap-in protrusion.

[0011] The snap-in protrusion is formed by a plurality of snap-in units spaced at intervals, the snap-in groove is formed by a plurality of snap-in groove units spaced at intervals, and the transverse protrusions are engaged with the transverse grooves after the plurality of snap-in units are engaged with the plurality of snap-in groove units one by one, to completely seal the mouth.

[0012] The plurality of snap-in units are buckles or hooks, the plurality of snap-in groove units are recesses configured to be detachably connected to the buckles or hooks, and the transverse protrusions are engaged with the transverse grooves after the buckles or hooks are connected to the recesses one by one, to completely seal the mouth.

[0013] Each of the buckles or hooks is T-shaped, and each of the recesses is formed as an inverted T-shaped counter bore or an inverted T-shaped groove.

[0014] The mouth is provided with a first abutment strip and a second abutment strip on two sides, the buckles or hooks are arranged on the first abutment strip and are spaced at intervals, a narrower end of each of the buckles or hooks is connected to the first abutment strip, a wider end of each of the buckles or hooks is away from the first abutment strip, the recesses are defined on the second abutment strip and spaced at intervals, a narrower end of the inverted T-shaped counter bore or the inverted T-shaped groove faces the first abutment strip, a wider end of the inverted T-shaped counter bore or inverted T-shaped groove is away from the first abutment strip, and when the mouth is completely sealed, the first abutment strip abuts against the second abutment strip.

[0015] A buffer strip is connected between two adjacent ones of the buckles or hooks, and the buffer strip protrudes from the first abutment strip.

[0016] The top of the body is configured with a lifting hole.

[0017] A first side edge of the body is connected with a hanging ear.

[0018] A manufacturing method of a sealing structure of the fresh-keeping bag is that the body, the sealing strips forming the mouth, and protrusions and grooves on the sealing strips are formed in a one-step in-mold forming process.

[0019] The fresh-keeping bag is made of silicone.

[0020] By adopting the above technical solution, the present application has the following beneficial effects.

[0021] The fresh-keeping bag and its mouth are molded in a one-step in-mold forming process, avoiding two processes of secondary manual adhesive process and solidifying by mold heating. The process is simplified, which, on one hand, reduces costs, and on the other hand, improves production efficiency and enhances the stability of quality, and additionally, which can also avoid the problem of “tearing open” in a bag mouth with conventional design.

[0022] The present application provides transverse protrusions between the two ends of the snap-in protrusion and the bottom of the mouth, and transverse grooves corresponding to the transverse protrusions between the two ends of the snap-in groove and the bottom of the mouth, to realize a complete seal of the fresh-keeping bag. A combination of vertical and horizontal arrangements can also ensure a stability of a seal.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0023] The present application is further described in detail below with reference to accompanying drawings.

[0024] FIG. 1 is a side view showing the structure of an existing fresh-keeping bag;

[0025] FIG. 2 is a partially enlarged side view of the existing fresh-keeping bag;

[0026] FIG. 3 is a left side view of a fresh-keeping bag of the present application;

[0027] FIG. 4 is a front view of the fresh-keeping bag of the present application;

[0028] FIG. 5 is a right side view of the fresh-keeping bag of the present application;

[0029] FIG. 6 is a perspective drawing of the fresh-keeping bag in the present application;

[0030] FIG. 7 is a cut-open view of the fresh-keeping bag of the present application;

[0031] FIG. 8 is an enlarged view of portion A in FIG. 7;

[0032] FIG. 9 is an enlarged view of portion B in FIG. 7;

[0033] FIG. 10 is a perspective drawing of the fresh-keeping bag of the present application from another perspective;

[0034] FIG. 11 is an enlarged view of portion C in FIG. 10;

[0035] FIG. 12 is a partially cross-sectional view of FIG. 10;

[0036] FIG. 13 is an enlarged view of portion D in FIG. 12;

[0037] FIG. 14 is a cross-sectional view of the fresh-keeping bag shown in FIG. 3;

[0038] FIG. 15 is an enlarged view of portion E in FIG. 14;

[0039] FIG. 16 is a perspective view of Embodiment 2 of the present application;

[0040] FIG. 17 is a partially enlarged view of 15F in FIG. 16 of the present application;

[0041] FIG. 18 is another perspective drawing of Embodiment 2 of the present application.

DETAILED DESCRIPTION

[0042] The specific embodiments of the present application are described below, which are only to make it easier for those skilled in the art to understand the technical solution of the present application, and not intended to limit the protection scope of the present application. Any equivalent change that can be obtained from the description of the embodiments of the present application should be within the protection scope of the present application.

Embodiment 1

[0043] Referring to FIGS. 3-15, a fresh-keeping bag of the present application is provided with a mouth **13** at a top of a body **10**, a first sealing strip **131** and a second sealing strip **132** are respectively provided on two sides of the mouth **13**. A snap-in protrusion **1311** is provided on the first sealing strip **131**, and a snap-in groove **1321** is defined on the second sealing strip **132** at a position corresponding to the snap-in protrusion **1311**. The snap-in protrusion **1311** can be embedded into the snap-in groove **1321**, to preliminarily seal the mouth **13**.

[0044] When users pinch the first sealing strip **131** and the second sealing strip **132** along an extension direction of the first sealing strip **131** and the second sealing strip **132** with their hands, the snap-in protrusion **1311** can be pressed into the snap-in groove **1321** to conveniently achieve a preliminary sealing effect.

[0045] It is difficult to achieve complete sealing between the snap-in protrusion **1311** and the snap-in groove **1321** at two ends thereof. In order to avoid this problem, transverse protrusions **1315** are provided between the two ends of the snap-in protrusion **1311** and a bottom of the mouth **13**, which extend horizontally relative to the snap-in protrusion **1311**, and transverse grooves **1325** are configured between the two ends of the snap-in groove **1321** and the bottom of the mouth **13** corresponding to the transverse protrusions **1315**. After the snap-in protrusion **1311** is engaged with the snap-in groove **1321**, the transverse protrusions **1315** are engaged with the transverse grooves **1325** to completely seal the mouth **13**.

[0046] The transverse grooves **1325** are designed as non-through in a thickness direction of the second sealing strip **132**, and groove ends are formed at two ends of each of the transverse grooves **1325** by the second sealing strip **132**. After the transverse protrusions **1315** are engaged with the transverse grooves **1325**, the groove ends are completely engaged with two ends of each of the transverse protrusions **1315**, which can ensure complete sealing of mouth **13**.

[0047] The transverse protrusion **1315** is semi-cylindrical in shape, which gradually rises from the bottom of the mouth **13** at a side of the transverse protrusion, and is connected with a curved transition to the snap-in protrusion **1311** at the other side of the transverse protrusion. A top of each of the transverse protrusions **1315** is higher than a top of the snap-in protrusion **1311**. Such a structure of the transverse protrusion **1315**, when engaged with a corresponding structure of the transverse groove **1325**, can achieve a smooth cooperation process between the transverse protrusion **1315** and the transverse groove **1325**.

[0048] A length of each of the transverse protrusions **1315** is greater than a thickness of the snap-in protrusion **1311**, and two transverse protrusions **1315** form a curved “I” in shape with the snap-in protrusion **1311**, such that a pulling force is generated when the transverse protrusions **1315** are engaged with the transverse grooves **1325**, and the snap-in protrusion **1311** is engaged with the snap-in groove **1321**, which increases a bonding force to tightly adhere a bottom of the first sealing strip **131** to a bottom the second sealing strip **132** (at a beginning of the mouth **13**), making it difficult to separate and achieving an effective seal.

[0049] In the sealing structure mentioned above, the body **10**, the first sealing strip **131** and the second sealing strip **132** forming the mouth **13**, as well as the snap-in protrusion **1311**, the transverse protrusions **1315**, the snap-in groove **1321**, and the transverse grooves **1325** on the first sealing strip **131** and the second sealing strip **132**, are manufactured using a one-step in-mold forming process, which effectively avoids defects caused by a secondary manual adhesive processing and solidifying by mold heating. The fresh-keeping bag of the present application is made of silicone, which fully utilizes characteristics of silicone while meeting the environmental requirements of the fresh-keeping bag, achieves one-step in-mold forming and the sealing performance of the sealing structure, and also meets requirements for repeated use.

Embodiment 2

[0050] Referring to FIGS. 3-15, the fresh-keeping bag of the present application is provided with a mouth **13** at the top of the body **10**, the first sealing strip **131** and the second sealing strip **132** are

provided on two sides of the mouth **13**. The snap-in protrusion **1311** is provided on the first sealing strip **131**, and the snap-in groove **1321** is formed on the second sealing strip **132** at the position corresponding to the snap-in protrusion **1311**. The snap-in protrusion **1311** can be embedded into the snap-in groove **1321**, thereby completing the preliminary sealing of the mouth **13**.

[0051] When users pinch the first sealing strip **131** and the second sealing strip **132** along an extension direction of the first sealing strip **131** and the second sealing strip **132** with their hands, the snap-in protrusion **1311** can be pressed into the snap-in groove **1321** to conveniently achieve a preliminary sealing effect.

[0052] It is difficult to achieve complete sealing between the snap-in protrusion **1311** and the snap-in groove **1321** at two ends thereof. In order to avoid this problem, transverse protrusions **1315** are provided between the two ends of the snap-in protrusion **1311** and a bottom of the mouth **13**, which extend horizontally relative to the snap-in protrusion **1311**, and transverse grooves **1325** are configured between the two ends of the snap-in groove **1321** and the bottom of the mouth **13** corresponding to the transverse protrusions **1315**. After the snap-in protrusion **1311** is engaged with the snap-in groove **1321**, the transverse protrusions **1315** are engaged with the transverse grooves **1325** to completely seal the mouth **13**.

[0053] An entire snap-in protrusion **1311** is engaged with an entire snap-in groove **1321**, a misalignment may occur during the engagement, which results in a certain section of the snap-in protrusion **1311** not being engaged with the corresponding section of the snap-in groove **1321**, the entire snap-in protrusion **1311** or a certain section of the entire snap-in groove **1321** may lift, forming a leakage port and thus leading to air leakage. The snap-in protrusion **1311** of Embodiment 2 is composed of a plurality of snap-in units **22** at intervals, and the snap-in groove **1321** is composed of a plurality of snap-in groove units **32** at intervals. After the plurality of snap-in units **22** are engaged with the plurality of snap-in groove units **32** one by one, the transverse protrusions **1315** are engaged with the transverse grooves **1325**, thereby completely sealing the mouth **13**. During a closure of the mouth **13**, the plurality of snap-in units **22** are engaged with the plurality of snap-in groove units **32** one by one, so there will be no misalignment, and the entire snap-in protrusion **1311** or the certain section of the entire snap-in groove **1321** will not lift.

[0054] The snap-in units **22** are buckles or hooks **221**, and the snap-in groove units **32** are recesses **321** that can be detachably connected to the buckles or hooks **221**. After a plurality of buckles or hooks **221** are connected to a plurality of recesses **321** one by one, the transverse protrusions **1315** are engaged with the transverse grooves **1325** to completely seal the mouth **13**. By the connection between the plurality of buckles or hooks **221** with the plurality of recesses **321** one by one, a stable connection between the buckles or hooks **221** and the recesses **321** can be achieved. When closing the mouth **13**, the plurality of buckles or hooks **221** on one side of the mouth **13** are engaged with the plurality of recesses **321** on the other side of the mouth **13** one by one, so it is less likely to cause misalignment, avoiding incomplete closure of the mouth **13** and potential air leakage port.

[0055] Each of the buckles or hooks **221** is T-shaped, and each of the recesses **321** is formed as an inverted T-shaped counter bore or an inverted T-shaped groove.

[0056] The mouth **13** is provided with a first abutment strip **21** and a second abutment strip **31** on two sides thereof, the plurality of buckles or hooks **221** are spaced at intervals along the first abutment strip **21**, and a narrower end of each of the buckles or hooks **221** is connected to the first abutment strip **21**, a wider end of each of the buckles or hooks **221** is away from the first abutment strip **21**. The plurality of recesses **321** are defined on the second abutment strip **31** and spaced at intervals, and a narrower end of the inverted T-shaped counter bore or the inverted T-shaped groove is close to the first abutment strip **21**, and a wider end of the inverted T-shaped counter bore or inverted T-shaped groove is away from the first abutment strip **21**. When the mouth **13** is completely sealed, the first abutment strip **21** abuts against the second abutment strip **31**, thereby improving the seal of the mouth **13**.

[0057] Specifically, the fresh-keeping bag is formed by one-piece thermal curing of silicone, so that both the buckles or hooks **221** and the recesses **321** have good elastic deformation performance. When the buckles or hooks **221** are engaged with the recesses **321**, a wider end edge of each of the buckles or hooks **221** is compressed to deform and become smaller, and a narrower end of each of the recesses **321** is pressed outward to be enlarged by the buckles or hooks **221**, therefore, the buckles or hooks **221** can be easily inserted into the recesses **321** to complete a clamping. Similarly, when it is necessary to open the mouth **13**, it can be realized by simply exerting force to pull apart two sides of mouth of the body **10**.

[0058] A buffer strip **23** is connected between two adjacent buckles or hooks **221**, and the buffer strip **23** protrudes from the first abutment strip **21**. Specifically, the buffer strip **23** is a silicone buffer strip, which has a certain degree of elastic deformation, thus playing a buffering role when the mouth **13** is closed while improving the seal of the mouth **13**.

[0059] The top of the body **10** is provided with a lifting hole **4** for hanging up the fresh-keeping bag.

[0060] One side edge of the body **10** is connected with a hanging ear **5** for hanging tags or labels.

Claims

1. A fresh-keeping bag, comprising a body, wherein a mouth is defined at a top of the body, a first sealing strip and a second sealing strip are respectively provided on two sides of the mouth, the first sealing strip is provided with a snap-in protrusion, a snap-in groove is defined on the second sealing strip at a position corresponding to the snap-in protrusion, the mouth is preliminarily sealed after the snap-in protrusion is engaged with the snap-in groove, transverse protrusions are provided between two ends of the snap-in protrusion and a bottom of the mouth, the transverse protrusions extend horizontally relative to the snap-in protrusion, transverse grooves corresponding to the transverse protrusions are formed between two ends of the snap-in groove and the bottom of the mouth, and the transverse protrusions are engaged with the transverse grooves after the snap-in protrusion is engaged with the snap-in groove, to completely seal the mouth.

2. The fresh-keeping bag according to claim 1, wherein each of the transverse grooves is designed as non-through in a thickness direction of first sealing strip and the second sealing strip, the first sealing strip and the second sealing strip respectively form groove ends at two ends of each of the transverse grooves, and the groove ends are completely engaged with two ends of each of the transverse protrusions after the transverse protrusions are engaged with the transverse grooves, to completely seal the mouth.

3. The fresh-keeping bag according to claim 1, wherein the transverse protrusions are semi-cylindrical in shape, a first side of each of the transverse protrusions gradually rises from the bottom of the mouth, and a second side of each of the transverse protrusions is connected with a curved transition to the snap-in protrusion, and a top of each of the transverse protrusions is higher than a top of the snap-in protrusion.

4. The fresh-keeping bag according to claim 2, wherein the transverse protrusions are semi-cylindrical in shape, a first side of each of the transverse protrusions gradually rises from the bottom of the mouth, a second side of each of the transverse protrusions is connected with a curved transition to the snap-in protrusion, and a top of each of the transverse protrusions is higher than a top of the snap-in protrusion.

5. The fresh-keeping bag according to claim 1, wherein a length of each of the transverse protrusions is greater than a thickness of the snap-in protrusion, and two of the transverse protrusions form a curved “I” in shape with the snap-in protrusion.

6. The fresh-keeping bag according to claim 2, wherein a length of each of the transverse protrusions is greater than a thickness of the snap-in protrusion, and two of the transverse protrusions form a curved “I” in shape with the snap-in protrusion.

7. The fresh-keeping bag according to claim 3, wherein a length of each of the transverse protrusions is greater than a thickness of the snap-in protrusion, and two of the transverse protrusions form a curved “I” in shape with the snap-in protrusion.
8. The fresh-keeping bag according to claim 4, wherein a length of each of the transverse protrusions is greater than a thickness of the snap-in protrusion, and two of the transverse protrusions form a curved “I” in shape with the snap-in protrusion.
9. The fresh-keeping bag according to claim 1, wherein the snap-in protrusion is formed by a plurality of snap-in units spaced at intervals, the snap-in groove is formed by a plurality of snap-in groove units spaced at intervals, and the transverse protrusions are engaged with the transverse grooves after the plurality of snap-in units are engaged with the plurality of snap-in groove units one by one, to completely seal the mouth.
10. The fresh-keeping bag according to claim 9, wherein the plurality of snap-in units are buckles or hooks, the plurality of snap-in groove units are recesses configured to be detachably connected to the buckles or the hooks, and the transverse protrusions are engaged with the transverse grooves after the buckles or the hooks are connected to the recesses one by one, to completely seal the mouth.
11. The fresh-keeping bag according to claim 10, wherein each of the buckles or the hooks is T-shaped, and each of the recesses is formed as an inverted T-shaped counter bore or an inverted T-shaped groove.
12. The fresh-keeping bag according to claim 11, wherein the mouth is provided with a first abutment strip and a second abutment strip on two sides, the buckles or the hooks are arranged on the first abutment strip and are spaced at intervals, a narrower end of each of the buckles or the hooks is connected to the first abutment strip, a wider end of each of the buckles or the hooks is away from the first abutment strip, the recesses are defined on the second abutment strip and spaced at intervals, a narrower end of the inverted T-shaped counter bore or the inverted T-shaped groove faces the first abutment strip, a wider end of the inverted T-shaped counter bore or the inverted T-shaped groove is away from the first abutment strip, and when the mouth is completely sealed, the first abutment strip abuts against the second abutment strip.
13. The fresh-keeping bag according to claim 12, wherein a buffer strip is connected between two adjacent ones of the buckles or the hooks, and the buffer strip protrudes from the first abutment strip.
14. The fresh-keeping bag according to claim 12, wherein the top of the body is configured with a lifting hole.
15. The fresh-keeping bag according to claim 12, wherein a first side edge of the body is connected with a hanging ear.
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